A-A-2235
September 26, 1985

COMMERICAL ITEM DESCRIPTION

CHAIRS, ROTARY, (OFFICE, ERGONOMIC, UPHOLSTERED)

The General Services Administration has authorized the use of this Commercial Item Description.

Salient characteristics:

Classification.

Type I - Task chair; nontilt seat; adjustable locking back tilt; adjustable locking seat height; casters.

Style A - With arms.
Style B - Without arms.

Size 1 - Small.
Size 2 - Medium.
Size 3 - Large.

Type II - Task chair; adjustable locking seat tilt; adjustable locking back tilt independent from seat tilt; adjustable locking seat height, casters

Style A - With arms.
Style B - Without arms.

Size 1 - Small.
Size 2 - Medium
Size 3 - Large.

FSC 7110

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Type III - Task chair; adjustable, locking, synchronized seat and back tilt; adjustable locking seat height; casters.

Style A - With arms.
Style B - Without arms.

Size 1 - Small.
Size 2 - Medium.
Size 3 - Large.

Type IV - Managers chair; tilt seat and back; adjustable tilt tension; adjustable locking seat height; casters.

Style A - With arms.

Size 3 - Large.

Type V - Side chair; nontilt seat and back, fixed seat height; glides.

Style A - With arms.
Style B - Without arms

Size 2 - Medium.

Design. The chairs shall be designed for heavy duty use. There shall not be any defects or deficiencies in the design or material that negatively affects appearance, serviceability or that could cause personal injury or damage to clothing. The seat and backrest shall be separate upholstered units connected together by a single column back upright. The back of the chair shall have a plastic outer shell colored within the range of black, dark gray and dark brown. The chairs shall have a "waterfall" edge seat front, lumbar support with a vertically convex and horizontally concave radius, rounded edges, rounded corners and comfortable seating. The seat and back shall not be severely contoured so as to restrict the occupant's movement. Pneumatic cylinders and other parts subject to wear shall be easily field replaceable. The chairs shall be capable of being professionally reupholstered.

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Dimensions. All dimensions are in inches unless otherwise indicated.

	Size 1	Size 2	Size 3
Seat height adjustment range, min. Types I, II, III and IV chairs. 1/	17 to 21	17 to 21	17 to 21
Seat height. Type V chair. 1/	17 - 18½	17 - 18½	17 - 18½
Seat width, min. 2/	17¾	17¾	20
Seat depth. 3/	15 - 17	15 - 17	16 - 18
Backrest height. 4/	10 - 13	13 - 16	18, min.
Backrest width, min. 5/	15	15	17
Lumbar support height. 6/	6 - 9	6 - 9	6 - 9
Lumbar suport width, min. 7/	12	12	14
Center of lumbar support above SRP. 3/	6 - 10	6 - 10	6 - 10
Width between armrests, min.	17¼	17¼	19½
Seat pan angle adjustment range, min. Types II and III chairs. 3/	0° to 7°	0° to 7°	0° to 7°
Seat pan angle. Types I, IV and V chairs 3/	0° - 10°	0° - 10°	0° - 10°
Seat back-to-pan included angle adjustment range, min. Types I, II and III chairs. 3/	90° to 105°	90° to 105°	90° to 105°
Seat back-to-pan included angle. Types IV and V chairs. 3/	90° - 105°	90° - 105°	90° - 105°
Back angle to seat angle synchronization ratio (deg.:deg.) Type III chair.	2:1 - 3:1	2:1 - 3:1	2:1 - 3:1

1/ Measured in accordance with Fig. 1.
2/ Measured from edge to edge of the seat at the spindle center.
3/ Measured in accordance with Fig. 2.
4/ Measured from the top edge to the bottom edge of the backrest along the centerline.
5/ Measured from edge to edge of the backrest at the center of lumbar support.
6/ Measured from top to bottom of the vertically convex lumbar support.
7/ Measured from side to side of the horizontally concave lumbar support.

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Adjustments. Adjustable locking seat and/or back tilt and seat height shall be capable of being fixed at any position within the range. The above adjustments shall be very easily accomplished and completely accessible from a normally seated position (semi-seated position for seat height adjustment) by push-button or push-lever operated controls. Controls that are awkward to operate such as handgrip tightening, lever tightening, screw adjusting, etc. are not acceptable, except when used for tilt tension adjustment or for adjustments other than those above. All adjustments shall not release or change unintentionally after being set. Tilt tension adjustment shall have a positive stop that will prevent the complete removal of tension from the unit. A top activated pneumatic cylinder shall be used to accomplish seat height adjustment.

Adjustment and maintenance instructions. Instructions shall be attached to the chair which clearly illustrates the operation of all chair adjustments and the maintenance procedures required of the chair.

Casters. The casters shall be single wheel with a friction ring stem. Wheel diameter shall be 2-1/2 ± 1/16 in. Tread material shall be rubber or plastic with a Shore durometer D scale hardness of 35 to 50 at 70°F. The casters shall have a finish compatible with the finish of the chair base.

Glides. The diameter shall be not less than 1 in. The glides shall have a finish compatible with the finish of the chair base. The glides shall be removable without the use of special tools.

Base. The chair base shall be five leg pedestal design. The material for the base shall be steel or aluminum which may be covered with plastic, except when base finish A below is used. The finish shall resist scuffing, abrasive wear and damage from impact. The base finish shall be either A or B below as specified in the contract or order

A Anodized aluminum or chromium plating
B Within the range of the colors black, dark gray and dark brown

Armrests. The armrests shall be relatively cushioned by a resilient material colored within the range of black, dark gray and dark brown. The armrests shall not be upholstered. The armrests shall be relatively flat. The armrests shall be designed not to interfere with arm movement during typing or prevent the user from moving-in close to the work surface.

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Padding. The seat and back padding shall be Grade HR-II molded or slab high-resilience polyurethane foam complying with the requirements of ASTM D 3770, Flexible Cellular Materials - High-Resilience Polyurethane Foam (HR) (most recent issue). The padding shall comply with requirements of BIFMA F-1-1978, First Generation Voluntary Upholstered Furniture Flammability Standard for Business and Institutional Markets (most recent issue). Sizes 1 and 2 chairs shall have the nominal thickness of the polyurethane foam in the seat not less than 1-1/2 in. and the nominal thickness of the polyurethane foam in the back not less than 1 in. Size 3 chairs shall have the nominal thickness of the polyurethane foam in the seat not less than 2 in. and the nominal thickness of the polyurethane foam in the back not less than 1-1/2 in. The padding shall be incapable of unintentional removal or displacement from its intended position on the chairs.

Upholstery fabric. The upholstery fabric shall be 100% flat nylon, dyed, plain weave, not less than 21 oz./linear yd. (based on 54 in. width) exclusive of back coating complying with the requirements in Table I of ASTM D 3597, Woven Upholstery Fabrics - Plain, Tufted or Flocked (most recent issue), except:

1. Breaking strength (load) shall be 250 lb., min. for warp and filling.
2. Tongue tear strength shall be 20 lb., min. for warp and filling.
3. Resistance to yarn slippage shall be 40 lb., min. for warp and filling.
4. Surface abrasion shall pass heavy-duty requirements.
5. Under "7.6 Colorfastness to water", delete Note 6 and paragraphs 7.6.2, 7.6.3 and 7.6.4

The fabric shall comply with the requirements of BIFMA F-1-1978, First Generation Voluntary Upholstered Furniture Flammability Standard for Business and Institutional Markets (most recent issue). The upholstery fabric shall be incapable of unintentional removal or displacement from its intended position on the chairs. The fabric colors shall be as specified in the contract or order.

Finish. All exposed steel shall have a corrosion resistant finish. All exposed aluminum shall have an anodized finish. The finish on all material shall be compatible with the overall finish appearance of the chair. Chairs having chromium plated components shall have those chromium plated

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components tested in accordance with ASTM B 117, Salt Spray (Fog) Testing (most recent issue) for a period of 48 hr. The plating on tested components shall not show visual evidence of peel, uneven color, powder film, pits or stain, except widely scattered spots of corrosion not exceeding 1/16 in. along the greatest dimension are acceptable.

Marking. The bottom of the chair seat shall be legibly and permanently marked with the following information: manufacturer's name or trademark, contract or order number, national stock number, month and year of manufacture, brief maintenance instructions, brief adjustment instructions and the name and address where replacement casters and pneumatic cylinder(s) of the type used on the chair can be obtained.

Performance. The chairs shall comply with the applicable requirements of ANSI/BIFMA X5.1-1985 General-Purpose Office Chairs - Tests (except where modified) as indicated below:

1. Section 5. Back Pull Test. Under "5.3 Test Procedures, paragraph 2)," add "Set seat height to highest position. Perform two separate tests on chairs with locking tilt adjustments. First, test with all tilt adjustments unlocked to allow free tilt to the stops and second, test with all tilt adjustments locked as described herein."
2. Section 8. Base Test.
3. Section 9. Drop Test.
4. Section 10. Swivel Cycling Test.
5. Section 12. Seating Impact Test - Cyclic. Under "12.3 Test Procedures," delete paragraph 3), and change paragraph 4) and Fig. 8 to a 2 in. drop height above unloaded seat surface. Bag weight shall be 125 lb.
6. Section 13. Stability Test.
7. Section 14. Arm Strength Test - Vertical.
8. Section 15. Arm Strength Test - Horizontal.
9. Section 16. Back Durability Test. Under "16.3 Test Procedures, paragraph 2)," add "Set seat height to highest position. Perform two separate tests on chairs with locking tilt adjustments. First, test with all tilt adjustments unlocked to allow free tilt to the stops and second, test with all tilt adjustments locked as described herein."
10. Section 18. Caster/Chair Base Durability Test. Perform test with obstacle layout for 36,000 cycles.

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Workmanship. The chairs shall be free of workmanship defects that negatively affect serviceability, appearance or that could cause personal injury or damage to clothing. The chairs shall be neatly upholstered. The finish shall be free of defects and areas of thin or no coverage that affects appearance.

Regulatory requirement. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes:

The commercial item certification clause is not required when this Commercial Item Description is used for procurement.

Purchasers shall specify and include the following information in the contract or order:

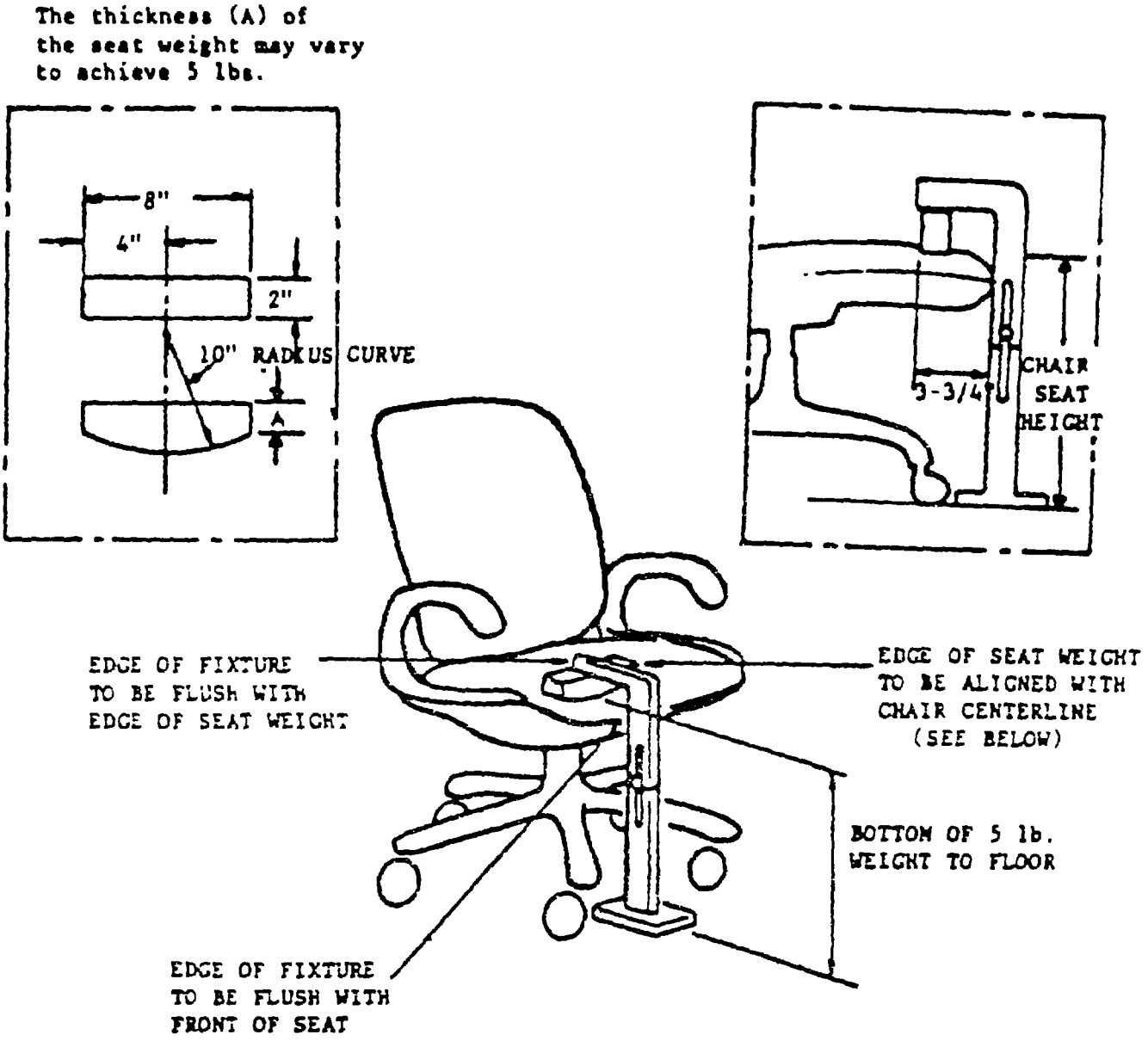
1. Type, Style and Size of chair desired (see Classification).
2. Base finish desired (see Base).
3. Color of upholstery fabric desired.
4. Warranty period required.

ANSI Standards are available from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

ASTM Standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

BIFMA Standards are available from The Business and Institutional Furniture Manufacturer's Association, 2335 Burton S.E., Grand Rapids, MI 49506.

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1. Place the chair on a level platform and adjust to the maximum height.
2. If the seat bar angle is adjustable, set to the minimum position but not less than zero degrees.
3. Weight the chair with 100 lbs placed so the weight does not compress the front of the seat cushion.
4. Place the center of a 5 lb. weight 4 in. to either the left or right of the chair centerline.
5. Measure from the top of the weight to the floor and subtract the thickness of the weight to obtain seat height.
6. Adjust the chair to the minimum height and repeat the measurement.

Fig. 1 Measurement technique for compressed seat height.

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1. Place chair on a level platform.

2. Place the mannequin measurement device onto the chair. Gently push device against seat and back surfaces to simulate the normal contact between a person and chair.

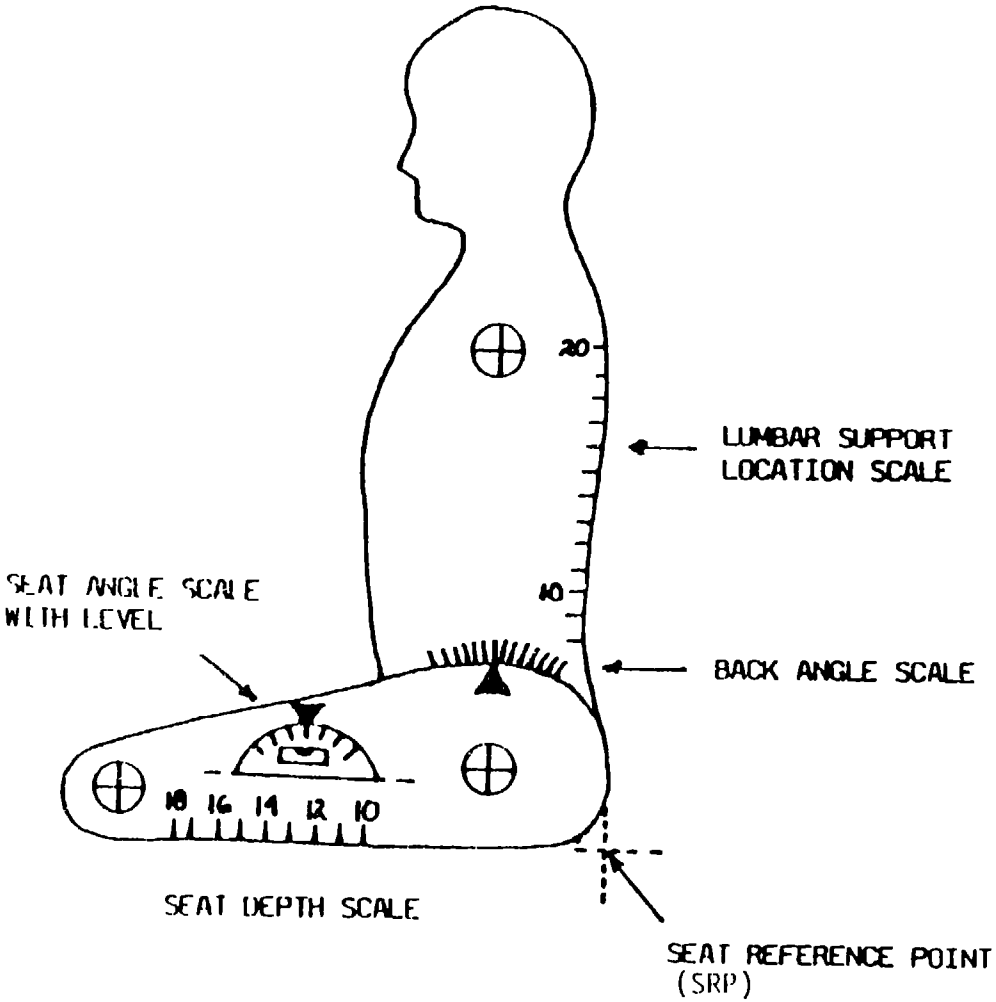
3. Set the seat angle measurement indicator to level.

4. Record the seat pan angle from the seat angle scale. If the seat angle is adjustable, set the seat to its minimum position (but not less than zero degrees) and record the seat angle.

5. Record the seat back-to-pan included angle from the back angle scale. If the seat back-to-pan included angle is adjustable, record the angle in both the minimum and maximum positions.

6. Record the seat depth from the seat depth scale.

7. Record the location of the center of the lumbar support from the lumbar support scale.



(See Fig 3 for design of mannequin measurement device.)

Fig. 2. Measurement technique for seat depth, seat pan angle, seat back-to-pan included angle, and center of lumbar support.

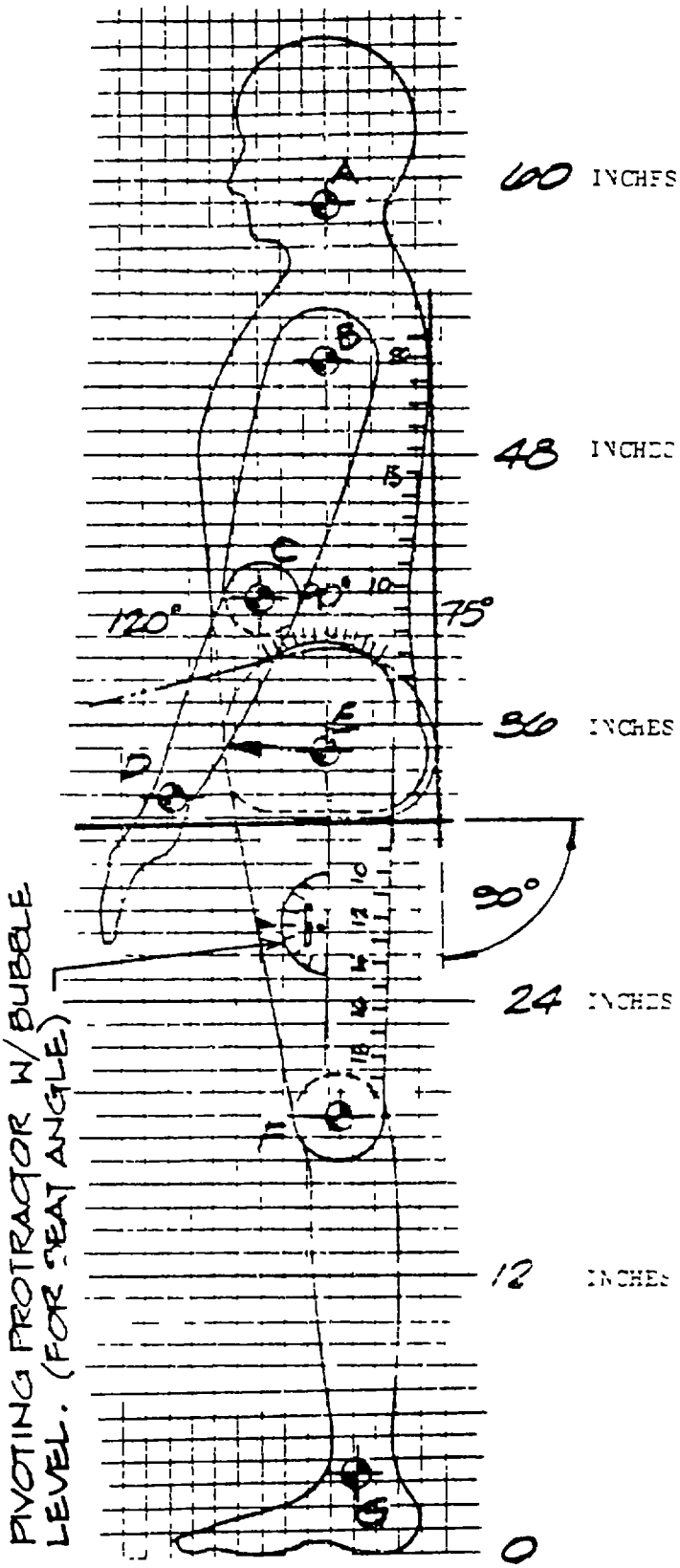


Fig. 3. Design of mannequin measurement device.

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